



Can auditing facilitate the effective implementation of goodwill accounting? A perspective based on key audit matters

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ABSTRACT

This study investigates the role of auditing in facilitating the effective implementation of goodwill accounting through key audit matters (KAMs). The findings show that excess goodwill increases the likelihood of goodwill being considered as a KAM, and firms with goodwill as a KAM are more likely to recognize goodwill impairment. The positive relationship between excess goodwill and a goodwill KAM is more pronounced when firms have a clean audit opinion, small profits before goodwill impairment, or high information transparency. However, the impact of a goodwill KAM on impairment recognition is weaker for firms with small profits. Overall, the findings highlight the crucial role of auditing in identifying clients' excess goodwill and encouraging the recognition of goodwill impairment, aligning with the perspectives of the International Accounting Standards Board.

1. Introduction

The debate on the “impairment-only approach” for subsequent goodwill measurement has been ongoing among stakeholders, academia, and regulatory authorities for a long time. Recently, in March 2024, the International Accounting Standards Board (IASB) published the “Exposure Draft *Business Combinations—Disclosures, Goodwill and Impairment*”,¹ proposing amendments to IFRS 3 *Business Combinations* and IAS 36 *Impairment of Assets*. These proposed changes to IAS 36 respond to stakeholders' concerns regarding untimely recognition of impairment losses on goodwill and the costly and complex nature of impairment tests. However, the IASB asserts that the issue of untimely impairment, partly stemming from management over-optimism, is best dealt with by auditors and enforcers rather than by amending IAS 36. In previous years, the IASB set up a research project titled “Goodwill and Impairment”. They gathered concerns from stakeholders regarding goodwill impairment and had already published a discussion paper in 2020,² posing the question “Can the impairment test be made more effective?” In response, they emphasized the critical role auditors play in addressing cases where firms delay goodwill impairment. In this context, it is crucial to investigate whether auditors can effectively identify issues with client goodwill impairment, thereby promoting the

effective implementation of goodwill standards.

Nevertheless, there is limited knowledge about the role auditors play in their clients' goodwill impairment recognition. Some studies address the impact of auditor characteristics (e.g., audit firm size, auditor industry specialization, auditor tenure) on the recognition and disclosure of goodwill impairment. They suggest that Big 4 auditors, auditor industry specialization, and auditor independence can enhance the compliance and timeliness of client goodwill impairment recognition and disclosure (Albersmann & Quick, 2020; Bepari & Mollik, 2015; Lobo et al., 2017). However, these studies do not probe into the audit process, leaving it a “black box” regarding how auditors detect and influence their clients' goodwill impairment behavior.

This study delves into the content of key audit matters (KAMs)³ and investigates whether audits can prompt goodwill impairment recognition by addressing it in KAMs in the Chinese context, thereby shedding light on the way audits influence the implementation of goodwill accounting. In alignment with international standards, the Ministry of Finance in China issued the “Chinese Audit Standard for Certified Public Accountants No. 1504” (CSACPA1504) in 2016, requiring listed firms' audit reports to include KAMs. This move aims to provide the market with more decision-useful information and enhance the value of audits. The new audit reporting standard was first applied to A + H listed firms

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¹ See the website: <https://www.ifrs.org/projects/work-plan/goodwill-and-impairment/exposure-draft-and-cl-bcdgi/>.

² See the website: <https://www.ifrs.org/content/dam/ifrs/project/goodwill-and-impairment/goodwill-and-impairment-dp-march-2020.pdf>.

³ The PCAOB refers to these matters as critical audit matters (CAMs). CAMs and KAMs are alternatives with the same meaning in this study.

from January 1, 2017, and extended to all A-share listed firms from January 1, 2018. KAMs outline those matters identified by the auditor as being of the most significance to the audit process. Their disclosure is akin to opening a window into the “black box” of auditing, allowing outsiders to understand the auditing process more clearly.

Specifically, this study empirically examines whether auditors can preemptively identify goodwill risks and subsequently prompt timely impairment recognition using a sample of A-share listed companies in China from 2017 to 2020. The results indicate that the greater the excess goodwill a firm possesses, the higher the probability that auditors include goodwill as a KAM. Moreover, firms for which goodwill is communicated as a KAM in the audit report are more likely to recognize goodwill impairments during the period, demonstrating the role of audits in prompting timely impairment recognition. These results are robust when we use the Propensity Score Matching (PSM) method, the fixed effects model, the change model, and the instrumental variable test to address endogeneity concerns, as well as when we use alternative proxies for key variables. Cross-sectional analyses show that the positive association between excess goodwill and a goodwill KAM is more pronounced when the firm receives a clean audit opinion, has small profits before goodwill impairment recognition, or has high information transparency, and the positive effect of a goodwill KAM on goodwill impairment recognition is mitigated for firms with small profits. The main effects hold in both large and small audit firms. Further analyses reveal that auditors exhibit little concern regarding “big bath” incentives using goodwill impairments. Additionally, firms with goodwill communicated as a KAM are more likely to recognize goodwill impairments in future periods and continuously reduce their excess goodwill.

This study makes several important contributions. Firstly, it responds to the recent call from IASB for comments on the amendments to IFRS 3 and IAS 36. The IASB has consistently asserted that management over-optimism, a main driver of the delays in recognizing impairment losses on goodwill, is best addressed by auditors and regulators rather than by changing IFRS accounting standards. Our findings support the IASB's assertion by demonstrating that auditors can indeed identify excess goodwill and facilitate the impairment of clients' goodwill. When developing accounting standards, standard-setters must balance the usefulness and relevance of the information produced with the associated costs and complexities. While the impairment-only approach is considered to provide more value-relevant information, it also brings greater costs, complexity, and increased opportunities for management manipulation, leading to criticisms of this method. However, our results suggest that the issue may not lie in the standards themselves but in the need for enhanced oversight during the implementation process. Given that “untimely impairment” is one of the stakeholders' major concerns regarding the impairment-only approach for subsequent goodwill measurement, our study underscores the role of auditors as independent external third parties in facilitating the effective implementation of goodwill standards, thus alleviating stakeholders' concerns to some certain extent.

Secondly, this study provides novel insights on how audits can perform an external governance role through KAM disclosures. Auditor behavior is difficult to observe, but KAMs offer a valuable lens into their actions and impacts. While goodwill impairment is a well-known issue within the audit profession and is frequently addressed through KAMs, limited knowledge exists on whether auditors can effectively identify excess goodwill in clients and substantially reduce the goodwill risk in financial reports. Our study delves into the content of KAMs to assess auditors' ability in identifying and disclosing goodwill risk, along with the corresponding actions taken by companies. We find that the likelihood of including goodwill in KAMs is significantly and positively associated with clients' excess goodwill, indicating that auditors are indeed capable of recognizing excess goodwill. We also find that firms with goodwill identified in KAM are more likely to recognize goodwill impairment, which means auditors do improve the quality of financial reports by KAM disclosure. These results offer a comprehensive view of

auditing governance role in improving the quality of financial reports and protecting the interests of relevant stakeholders by revealing risks beforehand and mitigating them afterward.

Thirdly, this study enhances the understanding of the practical impact of KAM disclosures on managerial behavior. The existing literature presents mixed conclusions on the effectiveness of KAM disclosures. While some studies document a significant impact of KAM disclosures on investor decisions, managerial risk-taking, financial reporting quality, and audit quality (Bentley et al., 2021; Christensen et al., 2014; Reid et al., 2019; Zeng et al., 2021), other studies find that KAM disclosures do not provide incremental information to the market nor influence audit fees or audit quality (Bédard et al., 2019; Burke et al., 2023). Our findings show that firms with goodwill as a KAM tend to exhibit a higher propensity to recognize goodwill impairment, establishing a positive link between auditor KAM identification and management's actual impairment recognition. This confirms the practical value of KAM disclosures and affirms the positive significance of audit standard reforms.

Fourthly, this study offers valuable perspectives for regulators and investors. Our findings show that auditors are more inclined to disclose goodwill risks in KAMs when they face higher potential risks. Notably, auditors tend to focus on identifying clients' excess goodwill while overlooking “big bath” behaviors related to goodwill impairment, which is another form of financial statement manipulation. These findings highlight auditors' incentives for risk management, emphasizing the need for regulators and investors to maintain strong oversight to ensure auditors fulfill their responsibilities effectively. Furthermore, our finding that KAM disclosures have predictive power over future managerial behavior offers important implications for investors, enabling them to make more informed decisions.

The remainder of this paper is as follows. Section 2 contains the literature review and hypotheses development. Section 3 discusses sample selection and research design. Section 4 reports the main results robustness checks. Section 5 provides additional analyses, and Section 6 concludes.

2. Literature review and hypotheses development

2.1. Auditing and goodwill impairment

Based on the agency theory, due to concerns about performance, compensation, reputation, violation of debt covenants, etc., managers tend to utilize the discretion granted by the goodwill impairment testing method to delay or avoid recognizing goodwill impairment (Beatty & Weber, 2006; Filip et al., 2021; Glaum et al., 2018; Li et al., 2011; Li & Sloan, 2017; Ramanna & Watts, 2012). As a result, the recognition of goodwill impairment usually lags behind deteriorating economic performance for several years (Albersmann & Quick, 2020; Hayn & Hughes, 2006; Jarva, 2009). According to IASB's report and discussion paper (International Accounting Standards Board (IASB), 2020; International Accounting Standards Board (IASB), 2024), impairment losses on goodwill are sometimes not recognized on a timely basis and the impairment test can be costly and complex are two main problems stakeholders proposed. The untimely recognition of impairment losses on goodwill not only distorts financial data and reduces the quality of accounting information but may also mislead investors, leading to temporarily inflated stock prices and future stock price collapses, and negatively affects a firm's performance growth (Han & Tang, 2020; Li & Sloan, 2017; Wu & Lai, 2020). Concerning these negative impacts, some studies have explored governance mechanisms for goodwill impairment, with some focusing on firms' internal governance and showing that firms with stronger corporate governance mechanisms are more likely to recognize goodwill impairment (Verriest & Gaeremynck, 2009), some others have also turned to external governance of firms, such as counties' enforcement intensity, monitoring through institutional investors, and analyst attention, which can promote timely goodwill

impairment recognition (Ayres et al., 2019; Glaum et al., 2018). Among these, several studies focus on the governance role of auditing in goodwill impairment.

Goodwill impairment tests frequently become the central concern for auditors, given the generally high risk of material misstatement and substantial scrutiny from regulatory bodies (Albersmann & Quick, 2020). Existing research generally supports the idea that high-quality audits can promote timely recognition of goodwill impairment. For instance, Big 4 auditors, whose audit quality is expected to be higher, lead to an increase in the timeliness of goodwill impairments and higher compliance with IFRS for goodwill impairment among their clients (Albersmann & Quick, 2020; Bepari & Mollik, 2015). Moreover, industry specialist auditors, expected to deliver high-quality audits as a result of greater competencies, can effectively monitor and encourage clients to report impairments promptly and record larger impairments (Stein, 2019). From the perspective of auditor independence, existing literature shows that the timeliness of goodwill impairment is negatively related to non-audit fees and auditor tenure (Albersmann & Quick, 2020; Carcello et al., 2020). Ayres et al. (2019) document that goodwill impairments are associated with a significant rise in the probability of auditor dismissals, indicating potential independence concerns as auditors strive to avoid dismissal by clients. In addition, Lobo et al. (2017) examine the effect of auditor pair composition on audit quality, documenting that firms audited by a Big 4–non-Big 4 auditor pair are more likely to recognize an impairment and book a larger impairment than firms audited by a Big 4–Big 4 auditor pair.

However, these studies do not investigate which audit procedures employed by auditors influence clients' goodwill impairment recognition behavior. Ma et al. (2024) examine the impact of mandatory KAM disclosure policy for A + H listed firms on goodwill impairment and investors' decision-making. They find that small audit firms fail to compel managers to promptly impair goodwill, contributing to further price distortion from goodwill inflation under the KAM reporting model. Their analysis concentrates on the impact derived from the expanded audit report without distinguishing whether the KAMs' content pertains to goodwill. Hence, their research falls short of shedding light on the auditor's capability to effectively discern the risk associated with goodwill impairment among clients. Consequently, how auditors play their role in clients' recognition of goodwill impairment remains unclear.

2.2. The consequences of disclosing KAMs

The implementation of auditing standards mandating the disclosure of KAMs in audit reports in major jurisdictions across the globe has triggered considerable debate regarding their utility for financial statements users. Recent studies, such as Smith (2023) and Seebeck and Kaya (2023), suggest that expanded audit reports with KAM disclosures enhance communicative value by improving readability and better reflecting firm characteristics and client-specific audit risks. From the perspective of investors and market reaction, prior research yields mixed results. Some studies demonstrate that KAM disclosures can influence investors' decisions, with market participants perceiving firms with extensive KAM/CAM disclosures as riskier (Christensen et al., 2014; Klevak et al., 2023). KAMs are also found to have an attention-directing effect on financial statement users (Sirois et al., 2018), and to affect investor perceptions of the value of the audit as well as the credibility of the auditor (Moroney et al., 2021). However, some studies find no evidence of market reactions to KAM disclosures, suggesting that investors may not perceive them as incrementally informative (Burke et al., 2023; Gutierrez et al., 2018). Rezaee and Homayoun (2024) conclude that except for severe KAMs, the audit regulation of disclosing KAM provides no value-relevant information to short sellers. Liao et al. (2023) find no significant association between auditor reporting of KAMs and stock price crash risk.

Several studies further explore the effects of KAM disclosures on the

debt market, auditor liability, financial reporting quality, and managerial risk-taking. Regarding the debt market, research shows that KAM disclosures can decrease interest rates and increase the proportion of long-term debt (Liu et al., 2022). Additionally, a higher number of disclosed KAMs correlates with greater financial distress (Camacho-Miñano et al., 2024). The role of KAMs in auditor liability is subject to debate, with conflicting perspectives offered by Brasel et al. (2016) and Gimbar et al. (2016). While the former suggests that CAM disclosures can reduce auditor liability judgments under certain conditions, the latter argues that the use of CAMs could increase auditor liability. Recent studies also investigate how KAM disclosures influence managerial behavior. Increased transparency through KAMs can lead to higher accountability pressure on managers and improve financial reporting quality (Gold et al., 2020). Burke et al. (2023) document significant changes to financial statement footnotes referenced by CAMs, implying potential shifts in management disclosure changes in areas expected to be scrutinized following auditor-provided disclosure. Bentley et al. (2021) examine the "Real Effects" of the CAM disclosures, finding that a CAM disclosure decreases managers' risk-decreasing activities more than managers' risk-increasing activities.

In summary, the current literature provides a nuanced view of the impacts of KAMs/CAMs across various stakeholder groups. However, these studies lack in-depth exploration of the specific content of KAMs and fail to provide evidence on whether auditors effectively identify and disclose clients' risks through KAMs. Additionally, research on the influence of KAM disclosures on managerial behavior overlooks the most direct actions linked to the contents of KAMs. For instance, goodwill is frequently cited as a KAM, but our understanding of how management responds to this is notably limited.

2.3. Hypotheses development

As noted by the IASB, the goodwill impairment test is complex, and impairment losses are sometimes recognized too late, raising concerns among stakeholders (International Accounting Standards Board (IASB), 2020; International Accounting Standards Board (IASB), 2024). Such delayed recognition of impairment can result in an overstatement of goodwill on firms' financial records (i.e., excess goodwill), and potentially leading to security mispricing and resource misallocation (Li & Sloan, 2017). The IASB identifies that untimely goodwill impairment recognition is mainly due to shielding and management over-optimism, which can be addressed by auditors. As gatekeepers of the capital market, auditors monitor management's accounting practices and convey their evaluation of the quality of accounting reports to investors through audit reports. Given that goodwill impairment relies heavily on managerial estimates and judgments, it is a critical area of auditor focus (Ma et al., 2024).

Under auditing standard CSACPA1504, which mandates a KAM section in audit reports, KAMs are intended to highlight issues that, in the auditor's professional judgment, hold significant importance in the audit process. For clients with substantial goodwill, auditors need to decide whether to identify goodwill as a KAM. Rational Choice Theory suggests that decision-makers evaluate the benefits and costs of each option, including both explicit and implicit costs. As rational economic agents, auditors weigh the benefits and costs when deciding whether to include goodwill as a KAM. An increase in excess goodwill amplifies audit risk, prompting auditors to allocate additional resources to detect and evaluate goodwill-related risks. By disclosing goodwill as a KAM, auditors not only enhance the transparency of financial reporting but also inform investors of heightened risks, thereby reducing the likelihood of audit failure. Such proactive disclosure also helps mitigate auditors' perceived liability. According to Brasel et al. (2016), when jurors are forewarned by KAM disclosures, they are more likely to perceive misstatements as foreseeable and less likely to hold auditors negligent in the event of plaintiff losses. This suggests that KAM disclosures can lower auditors' legal exposure, further incentivizing their use for clients

with significant excess goodwill. As illustrated in Fig. 1, the decision to identify and disclose goodwill as a KAM involves balancing associated costs—such as increased auditor effort and potential client complaints—against the benefits of reduced audit risk and perceived liability. As excess goodwill grows, the benefits of such disclosures are increasingly likely to outweigh the associated costs. Based on this analysis, we propose the first hypothesis:

H1. : The probability of auditors identifying goodwill as a KAM is positively correlated with the client's excess goodwill.

On the client side, the identification of goodwill as a KAM can influence management behavior regarding goodwill impairment. Management may overstate goodwill due to self-interest incentives, aiming to present more favorable financial results, or due to limited rationality and over-optimism. By identifying goodwill as a KAM, auditors can help mitigate opportunistic behaviors and irrational biases, promoting more timely recognition of impairment.

As illustrated in Fig. 1, when auditors identify goodwill as a KAM, clients may incur several increased potential costs. Firstly, auditors are likely to apply heightened scrutiny and perform additional audit procedures in this area. KAM disclosures require auditors to reference relevant financial statement accounts and provide reasonable justifications for identifying KAMs and their corresponding response measures. Details of audit work, previously concealed behind audit reports, are now compelled to be disclosed, enhancing the transparency of audit work and attracting external scrutiny. Under such pressure, auditors tend to exercise greater professional skepticism when auditing KAM matters. Consequently, when goodwill is identified as a KAM, indicating that auditors have invested additional time and resources in reviewing goodwill-related accounts, they are more likely to detect inflated goodwill and exert pressure on management. Given that managers tend to avoid aggressive reporting decisions when they are more easily to detect (Rose et al., 2013), they may adjust related estimation and recognize goodwill impairment more promptly. Furthermore, managers may anticipate that KAM disclosures will continue in future periods, creating a “threat of disclosure” (Reid et al., 2019) that encourages more conservative accounting.

Secondly, KAM disclosures attract greater attention from investors, regulators, and other stakeholders (Sirois et al., 2018), reducing the information asymmetry between management and external parties. When goodwill is identified as a KAM, management's aggressive

estimates in related accounts are more likely to be detected by financial statement users. This enhanced transparency may lead managers to fear being viewed as self-serving if they engage in aggressive estimation, risking reputational damage (Gold et al., 2020).

Therefore, when goodwill is identified as a KAM, the trade-off between the increased costs of untimely goodwill impairment recognition and the expected benefits is likely to lead clients to recognize goodwill impairment in a more timely manner. We propose the following hypothesis:

H2. : Firms for which goodwill is identified as a KAM are more likely to recognize goodwill impairment in the current period.

3. Sample selection and research design

3.1. Sample selection

CSACPA1504 requires all A-share listed companies to include the disclosure of KAMs in their audit reports for the 2017 financial statements and onward. Therefore, this study uses the period from 2017 to 2020 as the sample interval, with the initial sample being listed firms on the Shanghai and Shenzhen A-shares. We delete firms listed in the financial and insurance sectors as they differ significantly from those in other sectors. Additionally, firms under special treatment (i.e., those labeled ST or *ST), firms with missing relevant variables, and those with an initial goodwill net value of zero are excluded. The final sample consists of 6581 firm-year observations. All continuous variables are winsorized at the 1 % and 99 % levels. The financial and governance data are collected from the Wind database and the CSMAR (China Stock Market & Accounting Research) database.

3.2. Research design

3.2.1. Excess goodwill and the identification of goodwill as a KAM

To test whether auditors can effectively identify a firm's goodwill risk and disclose it as a KAM as excess goodwill increases (H1), we construct the following model:

$$KAM_{it} = \beta_0 + \beta_1 GW_{excess_{it-1}} + \beta_2 Controls_{it} + \sum Ind + \sum Year + \varepsilon_{it} \quad (1)$$

where KAM_{it} is an indicator variable that equals 1 if goodwill is included in KAMs of firm i in year t , and 0 otherwise. We extract KAMs of A-share listed firms from the Wind database and identify whether “goodwill” is mentioned in the content of KAMs. The independent variable $GW_{excess_{it-1}}$ denotes firms' excess goodwill at the end of last year. Following Luo and Guo (2023) and Wei and Zhu (2019), we use the regression residual of model (2) (a goodwill expectation model) as a proxy for excess goodwill.

$$GW_{it} = \beta_0 + \beta_1 GW_ind_{it} + \beta_2 CashPay_{it} + \beta_3 Size_{it} + \beta_4 Buyer_{it} + \beta_5 ROA_{it} + \beta_6 Dual_{it} + \beta_7 Growth_{it} + \beta_8 Mhold_{it} + \sum Ind + \sum Year + \varepsilon_{it} \quad (2)$$

GW_{it} is the firms' book goodwill standardized by total assets (goodwill-to-assets). GW_ind_{it} indicates the average goodwill-to-assets of other firms in the same industry for the same year. The model also includes other variables that may affect goodwill: whether the M&A transaction is paid for in cash ($CashPay_{it}$), the firm size ($Size_{it}$), buyers' expenditure ($Buyer_{it}$), the firm's profitability (ROA_{it}), the duality of CEO and chairman ($Dual_{it}$), the firms' growth capability ($Growth_{it}$), and the shareholding ratio of managers ($Mhold_{it}$). Year and industry dummy variables are also included in the model. The residual obtained from regression Model (2) represents the variance between actual goodwill and expected goodwill, serving as a metric for excess goodwill ($GW_{excess_{it}}$).

In model (1), we control for some firm characteristics that may affect

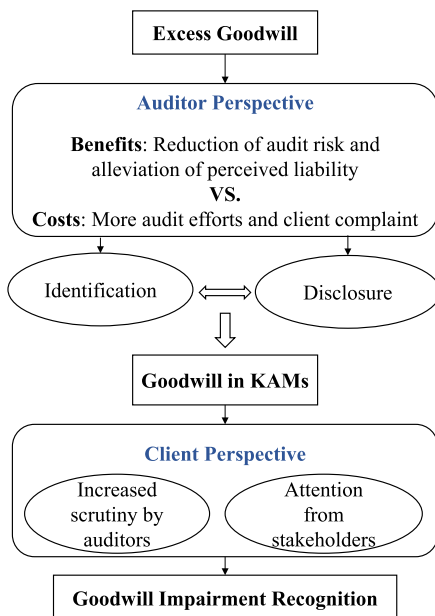


Fig. 1. The theoretical framework of the hypotheses.

goodwill KAM disclosures, such as the firm's size ($Size_{it}$), leverage ratio (Lev_{it}), profitability (ROA_{it}), growth capability ($Growth_{it}$), company age (Age_{it}), ownership concentration ($First_{it}$), duality of CEO and chairman ($Dual_{it}$), separation of ownership and control (Sep_{it}), net business credit (NTC_{it}), fixed asset ratio (PPE_{it}), and whether the firm is state-owned (SOE_{it}). Several auditing related variables are also controlled, such as whether the audit firm is an international Big 4 ($Big4_{it}$), audit fees ($Cost_{it}$), and audit opinion type ($Atype_{it}$). Additionally, industry (Ind) and year ($Year$) dummy variables are included in the regression. The specific definitions of these variables are detailed in [Appendix 1](#).

3.2.2. Identification of goodwill as a KAM and the firm's goodwill impairment

To test whether auditors' identification of goodwill in KAMs facilitates clients' recognition of goodwill impairment (H2), we develop the following model:

$$IMP_{it} = \beta_0 + \beta_1 KAM_{it} + \beta_2 Controls_{it} + \sum Ind + \sum Year + \varepsilon_{it} \quad (3)$$

where IMP_{it} is an indicator variable that equals 1 if goodwill is impaired for firm i in year t , and 0 otherwise. KAM_{it} indicates whether "goodwill" is identified as a KAM in the audit report, which is defined the same as that in model (1). Model (3) also includes control variables along with industry and year dummy variables.

4. Empirical results

4.1. Descriptive statistics

[Table 1](#) presents the descriptive statistics for the main variables. The mean value of KAM is 0.471, indicating that about 47 % of the observations have goodwill listed in KAMs in the audit reports. The mean value of the IMP is 0.367, indicating that 36.7 % of the firms conduct goodwill impairment during the sample period. The descriptive statistics of $Gwexcess$ are similar to those reported in [Luo and Guo \(2023\)](#). Notably, the mean value of $Gwexcess$ is larger than its median value, suggesting that the distribution of excess goodwill is skewed right and the overestimation of goodwill is quite serious for some firms in China. The distribution of other control variables is within a reasonable range. For example, the mean value of $Atype$ is 0.03, indicating that 3 % of the firms receive modified audit opinions, which is consistent with [Deng et al. \(2023\)](#).

4.2. Univariate tests for H1 and H2

We first conduct univariate tests for the two hypotheses. Specifically, for Hypothesis 1, we divide the sample into high and low groups based on whether excess goodwill exceeds the annual industry median. We conduct a t -test on the means of KAM between these two groups, and the results are presented in Panel A of [Table 2](#). The mean value of KAM for the low excess goodwill group is 0.2322, whereas for the high group, it is 0.7117. The significant difference indicates that firms with higher excess goodwill are more likely to attract audit attention, and auditors are more inclined to include goodwill in KAMs for these firms, aligning with Hypothesis 1. For Hypothesis 2, we examine the difference in goodwill impairment probability (IMP) between firms with different levels of audit attention on goodwill ($KAM = 1$ vs $KAM = 0$). The results in Panel B of [Table 2](#) show that firms with goodwill included in KAMs have a significantly higher probability of impairing goodwill, indicating that auditors' inclusion of goodwill in KAMs promotes firms to recognize goodwill impairment, consistent with Hypothesis 2. Furthermore, we divide the firms into two groups based on whether excess goodwill is greater than zero and conduct a chi-square test on the likelihood of including goodwill in KAMs between the groups. The results in Panel C of [Table 2](#) reveal that among 2256 observations with excess goodwill greater than zero, over 80 % have goodwill mentioned in KAMs. In

contrast, among the observations with excess goodwill less than or equal to zero, only about 30 % have goodwill as a KAM. These results provide initial evidence that auditors can, to some extent, identify goodwill risks and disclose them through KAMs.

4.3. Regression analyses for H1 and H2

[Table 3](#) reports the regression results for Hypothesis 1 and Hypothesis 2. Columns (1) and (2) show the logit regression results of model (1), with control variables omitted and included, respectively. The coefficients on $GWexcess$ are significantly positive at the 1 % level, indicating that the larger the excess goodwill of a firm, the more likely auditors are to communicate with external parties regarding goodwill through KAMs. These results support Hypothesis 1, suggesting that auditors can effectively identify excess goodwill and are motivated to disclose it in KAMs. Columns (3) and (4) present the logit regression results of model (3). The coefficients on KAM are significantly positive at the 1 % level, indicating that including goodwill in KAMs increases the probability of firms conducting impairment of goodwill. This suggests that auditors play a governance role by prompting management to recognize goodwill impairment in a timely manner, supporting Hypothesis 2.

4.4. Addressing the endogeneity concerns

There may exist endogeneity concerns regarding the above results. Although we have controlled lots of firm and auditing characteristics in the regression models, it is possible that other factors simultaneously affect both the independent and dependent variables. Firms with different levels of goodwill may inherently have significant differences at the corporate level, which may also influence auditors' judgment and decision on the content of KAMs. Therefore, our evidence on Hypothesis 1 may be a result of these inherent differences. Similarly, in Hypothesis 2, firms' decisions on goodwill impairment might also be influenced by inherent factors such as their operating conditions and management decision-making behavior.

To address this concern, firstly we use the Propensity Score Matching (PSM) method to make the sample firms more comparable. Specifically, to test Hypothesis 1, we divide the sample into two groups based on whether the excess goodwill exceeds the annual industry median. We then pair one high-excess goodwill group firm with the nearest-scored low-excess goodwill group firm on a one-to-one basis with no replacement.⁴ Based on the matched sample, we re-run model (1) and the results are shown in column (1) of [Table 4](#). The coefficient on $GWexcess$ is still significantly positive, supporting Hypothesis 1. Similarly, to test Hypothesis 2, we match each firm with goodwill in KAMs ($KAM = 1$) with the nearest-scored firm without goodwill in KAMs ($KAM = 0$) on a one-to-one basis, with no replacement.⁵ The regression results of model (3) for this subsample are reported in column (2) of [Table 4](#). The coefficient on KAM is still significantly positive, supporting Hypothesis 2.

Secondly, to control for the potential impact of time-invariant firm-level characteristics that may have been omitted, we further include firm fixed effects in models (1) and (3) to test Hypotheses 1 and 2. The results are shown in columns (3) and (4) of [Table 4](#). The coefficients on $GWexcess$ and KAM remain significantly positive in the fixed effects model.

Thirdly, we employ change model analyses to further address concerns of endogeneity. This approach mitigates potential biases in level regressions, which may result from unobserved, time-invariant variables. First, we regress the change in KAM (ΔKAM) on the change in $GWexcess$ ($\Delta GWexcess$), along with changes in the control variables. The regression results are reported in column (5) of [Table 4](#), where the

⁴ The variables used for matching are all the control variables in Model (1).

⁵ The variables used for matching are all the control variables in Model (3).

Table 1
Descriptive statistics.

Variables	Observations	Mean	Std. Dev.	Min	Max	Median
KAM_{it}	6581	0.471	0.499	0.000	1.000	0.000
IMP_{it}	6581	0.367	0.482	0.000	1.000	0.000
$GWexcess_{it-1}$	6581	0.002	0.092	-0.142	0.343	-0.023
$Size_{it}$	6581	22.640	1.266	20.400	26.610	22.440
Lev_{it}	6581	0.441	0.188	0.078	0.865	0.435
ROA_{it}	6581	0.028	0.082	-0.395	0.195	0.035
$Growth_{it}$	6581	0.146	0.315	-0.539	1.614	0.106
Age_{it}	6581	2.299	0.674	0.693	3.332	2.303
$First_{it}$	6581	31.657	13.770	8.920	70.760	29.490
$Dual_{it}$	6581	0.287	0.452	0.000	1.000	0.000
Sep_{it}	6581	0.042	0.070	0.000	0.279	0.000
NTC_{it}	6581	-0.093	0.296	-1.038	1.109	-0.080
PPE_{it}	6581	0.186	0.143	0.002	0.648	0.155
SOE_{it}	6581	0.303	0.459	0.000	1.000	0.000
$Big4_{it}$	6581	0.070	0.256	0.000	1.000	0.000
$Cost_{it}$	6581	14.130	0.695	12.900	16.690	14.000
$Atype_{it}$	6581	0.030	0.170	0.000	1.000	0.000

Table 1 presents the descriptive statistics for all the variables. All the variables are defined in Appendix 1.

Table 2
Univariate tests for H1 and H2.

	Lower excess goodwill (N = 3307)	Higher excess goodwill (N = 3274)	Difference (t-test)
Panel A: t-test on the means of KAM_{it}			
KAM_{it}	Mean 0.2322	Mean 0.7117	-0.4794***
Panel B: t-test on the means of IMP_{it}			
	$KAM_{it} = 0$ (N = 3483)	$KAM_{it} = 1$ (N = 3098)	Difference (T-test)
IMP_{it}	Mean 0.2461	Mean 0.5023	-0.2562***
Panel C: Chi-square test on the likelihood of KAM_{it}			
	$GWexcess_{it-1} \leq 0$	$GWexcess_{it-1} > 0$	Total
$KAM_{it} = 0$	3091 71.47 %	392 17.38 %	3483 52.93 %
$KAM_{it} = 1$	1234 28.53 %	1864 82.62 %	3098 47.07 %
Total	4325 100.00 %	2256 100.00 %	6581 100.00 %

Table 2 presents the univariate tests for H1 and H2. ***, **, and * indicate significance at 1 %, 5 %, and 10 %, respectively. Pearson chi2(1) = 1.7e+03 Pr = 0.000.

coefficient on $\Delta GWexcess$ is significantly positive. Next, we regress the change in IMP (ΔIMP) on the change in KAM (ΔKAM), as well as the change in the control variables. The results are shown in column (6) of Table 4, indicating a significant positively coefficient on ΔKAM . These results support our hypotheses.

Fourthly, we address the potential reverse causality issue associated with Hypothesis 2. When testing Hypothesis 1, the independent variable consists of lagged values, making the likelihood of reverse causality minimal. However, in the testing of Hypothesis 2, since the independent and dependent variables are contemporaneous, a potential reverse causality may exist. It is possible for a firm to recognize an impairment first, prompting the auditor to then include goodwill as a KAM. To address this concern, we use the instrumental variable method. Referring to existing literature (Luo & Guo, 2023), we utilize the industry average (KAM_mean_ind) and regional average (KAM_mean_region) of KAM as instrumental variables for KAM . According to social identity theory, imitation is quite common in business practice (Lieberman & Asaba, 2006). Auditors may refer to the disclosure of KAMs in the audit reports of other clients within the same industry or region as their own clients, and then adjust their disclosure of KAMs accordingly. Therefore,

Table 3
Regression analyses for H1 and H2.

	KAM_{it}		IMP_{it}	
	(1)	(2)	(3)	(4)
$GWexcess_{it-1}$	21.999*** (18.65)	21.335*** (16.89)		
KAM_{it}			1.122*** (16.59)	1.016*** (13.50)
$Size_{it}$		-0.205*** (-3.07)		0.108** (2.00)
Lev_{it}		0.199 (0.65)		-0.631** (-2.51)
ROA_{it}		-2.511*** (-3.39)		-9.301*** (-13.31)
$Growth_{it}$		0.580*** (4.70)		-0.449*** (-4.23)
Age_{it}		0.059 (0.75)		0.043 (0.65)
$First_{it}$		-0.013*** (-3.66)		0.006* (1.94)
$Dual_{it}$		0.335*** (3.46)		0.136* (1.74)
Sep_{it}		0.829 (1.31)		-0.723 (-1.39)
NTC_{it}		-0.487*** (-2.94)		-0.018 (-0.14)
PPE_{it}		-0.619* (-1.65)		-0.634** (-2.03)
SOE_{it}		-0.242** (-2.06)		-0.136 (-1.39)
$Big4_{it}$		0.262 (1.30)		-0.868*** (-4.97)
$Cost_{it}$		0.399*** (3.80)		0.206** (2.40)
$Atype_{it}$		-0.175 (-0.80)		0.058 (0.28)
Year	Yes	Yes	Yes	Yes
Ind	Yes	Yes	Yes	Yes
N	6581	6581	6581	6581
Pseudo R ²	0.267	0.292	0.071	0.140

Table 3 presents the regression results for H1 and H2. The figures in bold are the variables of our interest. This table presents the coefficients and z-values. ***, **, and * indicate significance at 1 %, 5 %, and 10 %, respectively.

KAM_mean_ind and KAM_mean_region are expected to be positively associated with KAM , while their direct effect on IMP is not anticipated. The two-stage regression results with KAM_mean_ind and KAM_mean_region as the instrumental variables are shown in Table 4, columns (7) and (8). In column (7), the coefficients on KAM_mean_ind and KAM_mean_region are significantly positive, consistent with our expectation. In

Table 4
Addressing the endogeneity concerns.

	PSM		Firm fixed effects		Change model		Instrumental variable	
	<i>KAM_{it}</i>	<i>IMP_{it}</i>	<i>KAM_{it}</i>	<i>IMP_{it}</i>	ΔKAM_{it}	ΔIMP_{it}	<i>KAM_{it}</i>	<i>IMP_{it}</i>
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>GWexcess_{it-1}</i> ($\Delta GWexcess_{it-1}$)	21.703*** (15.72)		8.519*** (5.78)		2.571*** (2.91)			
<i>KAM_mean_ind_{it}</i>							0.889*** (5.16)	
<i>KAM_mean_region_{it}</i>							0.836*** (17.07)	
<i>KAM_{it}</i> (ΔKAM_{it})		1.024*** (12.63)		1.052*** (7.06)		0.854*** (7.50)		
<i>KAM_p_{it}</i>								0.871*** (5.22)
<i>Size_{it}</i>	−0.224*** (−3.19)	0.115* (1.83)	2.031*** (5.47)	0.834*** (2.85)	1.458*** (5.25)	0.122 (0.69)	−0.036*** (−4.20)	0.070*** (2.64)
<i>Lev_{it}</i>	0.321 (0.97)	−0.644** (−2.22)	−0.606 (−0.56)	−0.072 (−0.09)	−0.192 (−0.24)	−0.679 (−1.28)	−0.191*** (−4.80)	−0.294** (−2.29)
<i>ROA_{it}</i>	−2.544*** (−2.89)	−9.375*** (−10.40)	−4.787*** (−3.87)	−15.063*** (−11.15)	−2.720*** (−4.19)	−5.947*** (−13.28)	−0.920*** (−11.24)	−5.022*** (−13.90)
<i>Growth_{it}</i>	0.646*** (4.75)	−0.200 (−1.50)	−0.080 (−0.37)	−0.188 (−1.13)	0.003 (0.02)	−0.100 (−0.95)	0.136*** (6.96)	−0.301*** (−4.73)
<i>Age_{it}</i>	−0.045 (−0.53)	0.109 (1.42)	0.109 (0.15)	0.334 (0.64)	—	—	0.036*** (3.38)	0.023 (0.70)
<i>First_{it}</i>	−0.014*** (−3.71)	0.007** (2.12)	−0.005 (−0.24)	−0.007 (−0.44)	−0.009 (−0.65)	−0.005 (−0.57)	−0.004*** (−7.79)	0.004*** (2.76)
<i>Dual_{it}</i>	0.339*** (3.34)	0.072 (0.82)	0.404* (1.72)	0.117 (0.67)	0.098 (0.55)	0.104 (0.84)	0.026* (1.94)	0.077* (1.92)
<i>Sep_{it}</i>	0.334 (0.49)	−0.884 (−1.49)	0.501 (0.20)	0.071 (0.04)	1.122 (0.84)	−0.523 (−0.45)	0.185** (2.21)	−0.477* (−1.87)
<i>NTC_{it}</i>	−0.518*** (−2.92)	−0.039 (−0.25)	−0.804* (−1.77)	0.105 (0.37)	−0.359 (−1.52)	−0.137 (−0.74)	−0.126*** (−5.55)	0.025 (0.34)
<i>PPE_{it}</i>	−0.261 (−0.65)	−0.534 (−1.44)	1.624 (0.92)	−0.739 (−0.52)	0.342 (0.30)	−0.865 (−1.08)	−0.439*** (−9.07)	−0.260 (−1.55)
<i>SOE_{it}</i>	−0.233* (−1.86)	−0.125 (−1.13)	−0.126 (−0.24)	0.018 (0.05)	0.333 (0.96)	0.192 (0.81)	−0.097*** (−6.39)	−0.059 (−1.20)
<i>Big4_{it}</i>	0.219 (1.02)	−0.836*** (−4.21)	0.446 (0.57)	−1.121 (−1.45)	0.523 (0.67)	0.053 (0.20)	0.050* (1.95)	−0.527*** (−6.52)
<i>Cost_{it}</i>	0.463*** (4.21)	0.127 (1.31)	0.101 (0.25)	0.227 (0.80)	−0.068 (−0.23)	−0.096 (−0.49)	0.111*** (8.14)	0.092** (2.02)
<i>Atype_{it}</i>	−0.135 (−0.51)	0.042 (0.18)	−0.235 (−0.60)	0.177 (0.51)	−0.129 (−0.41)	0.048 (0.24)	−0.064* (−1.85)	0.057 (0.53)
<i>Year</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Ind</i>	Yes	Yes	No	No	Yes	Yes	Yes	Yes
<i>Firm</i>	No	No	Yes	Yes	No	No	No	No
<i>N</i>	5092	4504	1698	3554	4506	4506	6581	6581
<i>Pseudo R²</i>	0.266	0.102	0.233	0.231	0.070	0.075	0.148	0.180

Table 4 presents the result of endogeneity tests for H1 and H2. The figures in bold are the variables of our interest. In the change model analysis in columns (5) and (6), all control variables are measured as the change from the previous period to the current period. For brevity, the notation for these control variables is not distinguished from that in other columns. The number of observations in columns (3) and (4) is lower than in other columns because firms whose dependent variable values did not change during the sample period are omitted in logit regressions when controlling for firm fixed effects. This table presents the coefficients and z-values or t-values. ***, **, and * indicate significance at 1 %, 5 %, and 10 %, respectively.

the second-stage regression (column (8)), the coefficient on the predicted *KAM* obtained from the first stage (*KAM_p*) is still significantly positive, indicating that our evidence on Hypothesis 2 is not affected by the reverse causality problem.

Overall, these analyses show that our results are not driven by endogeneity issues.

4.5. Alternative proxies for key variables

To enhance the robustness of the results, we employ a variety of alternative proxies for the key variables. Regarding excess goodwill, acknowledging that goodwill is affected by peer effect at the industry level (Wang & Huang, 2019), we draw on Ramanna (2008)'s conceptualization of abnormal goodwill to estimate excess goodwill. Specifically, we calculate the difference between the firm's goodwill-to-assets and the industry median (*GWexcess1*), as well as the difference between the firm's goodwill-to-assets and the industry average (*GWexcess2*) for the corresponding year. Further, following Li and Sloan (2017), we

introduce the variable *IMPI* as another alternative indicator for excess goodwill. If the firm's goodwill-to-assets exceed 10 % and *ROA* is less than zero, *IMPI* equals 1; if the firm's goodwill-to-assets is less than 5 % and *ROA* is greater than 5 %, *IMPI* equals −1; otherwise, *IMPI* equals 0. A higher value of *IMPI* signals a higher probability that goodwill should have been materially impaired, thus indicating a greater goodwill risk associated with the firm's goodwill. The regression results of model (1), utilizing *GWexcess1*, *GWexcess2*, and *IMPI* as alternative proxies for excess goodwill, are reported in columns (1), (2), and (3) of Table 5, respectively. The positive and statistically significant coefficients of these variables corroborate Hypothesis 1.

Moreover, in the primary regression analyses, we assess the auditor's attention to goodwill risk by determining whether "goodwill" is mentioned in the content of KAMs (*KAM*). In this section, we employ a more specific method of measurement: identifying whether "goodwill impairment" is listed as a KAM (*KAM1*). *KAM1* equals 1 if "goodwill impairment" is mentioned in the firm's KAMs, and 0 otherwise. We re-run both models (1) and (3) using *KAM1* as an alternative proxy for

Table 5

Alternative proxies for key variables.

	<i>KAM_{it}</i>			<i>KAM1_{it}</i>	<i>IMP_{it}</i>
	(1)	(2)	(3)	(4)	(5)
<i>GWexcess1_{it-1}</i>	25.395*** (14.81)				
<i>GWexcess2_{it-1}</i>		25.060*** (15.05)			
<i>IMPI_{it-1}</i>			1.473*** (14.71)		
<i>GWexcess_{it-1}</i>				19.224*** (16.09)	
<i>KAM1_{it}</i>					1.031*** (13.68)
<i>Size_{it}</i>	−0.025 (−0.38)	−0.027 (−0.41)	−0.069 (−1.05)	−0.193*** (−2.94)	0.107** (1.98)
<i>Lev_{it}</i>	−0.148 (−0.48)	−0.158 (−0.51)	−1.760*** (−5.99)	0.204 (0.67)	−0.623** (−2.49)
<i>ROA_{it}</i>	−1.953** (−2.57)	−1.972*** (−2.61)	−2.684*** (−5.25)	−2.291*** (−3.18)	−9.286*** (−13.29)
<i>Growth_{it}</i>	0.208* (1.70)	0.219* (1.79)	0.460*** (4.32)	0.487*** (4.13)	−0.443*** (−4.17)
<i>Age_{it}</i>	0.069 (0.87)	0.069 (0.87)	−0.032 (−0.40)	0.026 (0.34)	0.048 (0.73)
<i>First_{it}</i>	−0.008** (−2.27)	−0.008** (−2.27)	−0.014*** (−3.82)	−0.014*** (−3.89)	0.006** (2.01)
<i>Dual_{it}</i>	0.178* (1.83)	0.179* (1.84)	0.181* (1.95)	0.263*** (2.70)	0.144* (1.85)
<i>Sep_{it}</i>	1.007 (1.61)	0.972 (1.55)	1.111* (1.67)	0.997 (1.59)	−0.763 (−1.47)
<i>NTC_{it}</i>	−0.391** (−2.36)	−0.402** (−2.41)	−0.590*** (−3.58)	−0.427*** (−2.64)	−0.025 (−0.20)
<i>PPE_{it}</i>	−0.190 (−0.52)	−0.192 (−0.52)	−2.172*** (−5.53)	−0.573 (−1.54)	−0.639** (−2.04)
<i>SOE_{it}</i>	−0.068 (−0.58)	−0.072 (−0.61)	−0.472*** (−3.90)	−0.236** (−2.05)	−0.136 (−1.39)
<i>Big4_{it}</i>	0.317 (1.61)	0.316 (1.60)	0.295 (1.45)	0.219 (1.08)	−0.860*** (−4.94)
<i>Cost_{it}</i>	0.304*** (2.91)	0.309*** (2.96)	0.521*** (5.14)	0.362*** (3.46)	0.210** (2.46)
<i>Atype_{it}</i>	−0.255 (−1.10)	−0.256 (−1.10)	−0.276 (−1.45)	−0.272 (−1.21)	0.075 (0.36)
<i>Year</i>	Yes	Yes	Yes	Yes	Yes
<i>Ind</i>	Yes	Yes	Yes	Yes	Yes
<i>N</i>	6581	6581	6581	6581	6581
<i>Pseudo R²</i>	0.328	0.327	0.160	0.272	0.141

Table 5 presents the regression results for H1 and H2 after replacing key variables with alternative proxies. The figures in bold are the variables of our interest. This table presents the coefficients and z-values. ***, **, and * indicate significance at 1 %, 5 %, and 10 %, respectively.

KAM, and the results are shown in Table 5, columns (4) and (5). The coefficient on *GWexcess* in column (4) with *KAM1* as the dependent variable is significantly positive, supporting Hypothesis 1. The coefficient on *KAM1* in column (5) is also significantly positive, supporting Hypothesis 2. Overall, our results remain robust with alternative proxies for key variables.

5. Additional analyses

5.1. Cross sectional analyses

5.1.1. Audit opinion

Hypothesis 1 indicates that the larger the excess goodwill from the previous period, the more likely auditors are to mention goodwill in KAMs due to risk considerations. In auditing practice, if auditors find that a client's risks exceed acceptable limits, they may issue a modified opinion, which serves as a stronger warning to investors about these risks. In such cases, the role of KAMs as a reminder to investors decreases. Conversely, when auditors issue a clean opinion, they face stronger litigation risks from potential audit failures, and the role of KAMs in alerting investors about a client's risk becomes more pronounced. Therefore, it can be expected that the positive association between clients' excess goodwill and the probability of mentioning

goodwill in KAMs is more pronounced when auditors issue a clean opinion. Regarding Hypothesis 2, regardless of whether the auditor issues a modified opinion or a clean opinion, including goodwill in the KAMs indicates that the auditor has paid special attention to goodwill during the audit and performed corresponding audit procedures. This disclosure increases the transparency of the financial statements, allowing investors and regulators to understand the goodwill risks and putting pressure on management. Therefore, the positive relationship between the disclosure of goodwill in KAMs and the firm's recognition of goodwill impairment will exist regardless of the type of audit opinion.

We examine the moderating effect of audit opinion on our main hypotheses by dividing the sample into two groups based on whether the firm receives a modified opinion. The subsample regression results are reported in Table 6. Columns (1) and (2) show that the positive effect of excess goodwill on the probability of including goodwill in KAMs exists in both groups. However, the magnitude of the coefficient on *GWexcess* in the clean audit opinion group is significantly larger than in the modified audit opinion group, indicating a more pronounced effect in the former. Columns (3) and (4) show that the positive effect of a goodwill KAM on the probability of goodwill impairment exists in both groups, with no significant difference between the two groups. These results align with our expectations.

Table 6

Cross sectional analyses: Audit opinion.

	<i>KAM_{it}</i>		<i>IMP_{it}</i>	
	(1)	(2)	(3)	(4)
	Clean audit opinion	Modified audit opinion	Clean audit opinion	Modified audit opinion
<i>GW_{excessit-1}</i>	22.009*** (29.20)	12.809*** (4.40)		
<i>KAM_{it}</i>			1.031*** (13.45)	0.828** (2.01)
<i>Size_{it}</i>	−0.211*** (−4.36)	−0.120 (−0.41)	0.102* (1.85)	0.357 (1.05)
<i>Lev_{it}</i>	0.209 (0.89)	−1.123 (−1.03)	−0.641** (−2.52)	0.806 (0.66)
<i>ROA_{it}</i>	−3.080*** (−4.72)	1.526 (0.88)	−9.434*** (−12.90)	−9.552*** (−4.79)
<i>Growth_{it}</i>	0.645*** (5.58)	−0.277 (−0.52)	−0.511*** (−4.66)	0.365 (0.65)
<i>Age_{it}</i>	0.057 (0.98)	−0.022 (−0.05)	0.045 (0.67)	−0.689 (−1.48)
<i>First_{it}</i>	−0.012*** (−4.79)	−0.075*** (−3.09)	0.006** (1.99)	−0.024 (−0.95)
<i>Dual_{it}</i>	0.326*** (4.37)	0.578 (1.23)	0.152* (1.92)	−0.310 (−0.63)
<i>Sep_{it}</i>	0.842* (1.87)	5.178 (1.51)	−0.740 (−1.40)	−0.249 (−0.07)
<i>NTC_{it}</i>	−0.470*** (−3.58)	−0.242 (−0.43)	0.016 (0.12)	−1.000 (−1.45)
<i>PPE_{it}</i>	−0.596** (−2.26)	1.997 (1.07)	−0.676** (−2.15)	−1.198 (−0.53)
<i>SOE_{it}</i>	−0.246*** (−2.96)	1.028 (1.34)	−0.140 (−1.41)	−1.380* (−1.79)
<i>Big4_{it}</i>	0.277** (2.01)	−0.852 (−0.69)	−0.874*** (−4.93)	−0.270 (−0.21)
<i>Cost_{it}</i>	0.389*** (5.09)	1.285** (2.51)	0.217** (2.51)	−0.541 (−1.28)
<i>Year</i>	Yes	Yes	Yes	Yes
<i>Ind</i>	Yes	Yes	Yes	Yes
<i>N</i>	6384	184	6384	191
<i>Pseudo R²</i>	0.296	0.310	0.135	0.314
<i>Suest test (Chi²)</i>	4.55**		0.24	

Table 6 presents the cross sectional analysis results for H1 and H2 distinguishing the type of audit opinions. The figures in bold are the variables of our interest. This table presents the coefficients and z-values. ***, **, and * indicate significance at 1 %, 5 %, and 10 %, respectively.

5.1.2. Firms' small profits

We further consider the heterogeneity of Hypotheses 1 and 2 among clients with different characteristics. Firstly, we believe that firms' small profits before recognizing goodwill impairment can influence auditors' decisions and management's actions. Marginally profitable firms generally face greater performance pressure and delisting risks, leading to a higher risk of material misstatement. Therefore, auditors may be more cautious and more likely to disclose goodwill risks in KAMs when they have excess goodwill. From the clients' perspective, for marginally profitable firms, recognizing goodwill impairment could likely result in losses in profits, leading to more severe economic consequences. Thus, even under the pressure from auditors disclosing goodwill risks in KAMs, management may be reluctant to recognize goodwill impairment.

We conducted a subgroup analysis for firms with small profits and those without small profits before recognizing goodwill impairment, with results shown in **Table 7**. In columns (1) and (2), we find a significant positive relationship between clients' excess goodwill and the probability of auditors including goodwill in KAMs in both groups. However, the coefficient is significantly larger for firms with small profits, indicating that auditors are more inclined to disclose goodwill in KAMs when they are exposed to higher potential risks. This is consistent with our theoretical discussion of Hypothesis 1. According to columns (3) and (4), we find that the positive relationship between a goodwill KAM and goodwill impairment is significant in non-small profit firms.

Table 7

Cross sectional analyses: Firms' small profits.

	<i>KAM_{it}</i>		<i>IMP_{it}</i>	
	(1)	(2)	(3)	(4)
	Non-small profits	Small profits	Non-small profits	Small profits
<i>GW_{excessit-1}</i>	20.659*** (16.31)	29.595*** (5.84)		
<i>KAM_{it}</i>			1.074*** (13.56)	0.216 (1.03)
<i>Size_{it}</i>	−0.217*** (−3.16)	−0.200 (−1.03)	0.102* (1.80)	0.050 (0.34)
<i>Lev_{it}</i>	0.352 (1.10)	−1.174 (−1.37)	−0.625** (−2.34)	−1.836*** (−2.59)
<i>ROA_{it}</i>	−2.448*** (−3.28)	−47.683*** (−2.58)	−9.379*** (−12.92)	−120.532*** (−6.72)
<i>Growth_{it}</i>	0.568*** (4.40)	0.705 (1.53)	−0.489*** (−4.30)	−0.057 (−0.16)
<i>Age_{it}</i>	0.073 (0.92)	−0.116 (−0.43)	0.083 (1.21)	−0.329 (−1.44)
<i>First_{it}</i>	−0.012*** (−3.28)	−0.020** (−2.27)	0.007** (2.35)	−0.011 (−1.31)
<i>Dual_{it}</i>	0.317*** (3.12)	0.453* (1.84)	0.159* (1.92)	0.004 (0.02)
<i>Sep_{it}</i>	0.585 (0.87)	3.411** (2.12)	−0.831 (−1.51)	0.440 (0.29)
<i>NTC_{it}</i>	−0.495*** (−2.86)	−0.684 (−1.48)	−0.036 (−0.26)	−0.173 (−0.56)
<i>PPE_{it}</i>	−0.597 (−1.52)	−0.271 (−0.27)	−0.398 (−1.21)	−1.728** (−2.06)
<i>SOE_{it}</i>	−0.248** (−2.03)	−0.117 (−0.39)	−0.150 (−1.44)	0.247 (0.97)
<i>Big4_{it}</i>	0.272 (1.30)	0.036 (0.06)	−0.826*** (−4.59)	−1.853** (−2.51)
<i>Cost_{it}</i>	0.414*** (3.81)	0.515 (1.61)	0.211** (2.33)	0.374 (1.54)
<i>Atype_{it}</i>	−0.289 (−1.22)	1.132 (1.52)	0.113 (0.52)	−0.108 (−0.20)
<i>Year</i>	Yes	Yes	Yes	Yes
<i>Ind</i>	Yes	Yes	Yes	Yes
<i>N</i>	5900	664	5900	672
<i>Pseudo R²</i>	0.289	0.356	0.148	0.189
<i>Suest test (Chi²)</i>	3.05*		15.44***	

Table 7 presents the cross sectional analysis results for H1 and H2 distinguishing the existence of small profits. The figures in bold are the variables of our interest. This table presents the coefficients and z-values. ***, **, and * indicate significance at 1 %, 5 %, and 10 %, respectively.

However, in firms with small profits, auditors' attention to goodwill does not promote impairment recognition. This suggests that auditors encounter substantial resistance in prompting goodwill impairment in those firms.

5.1.3. Firms' information transparency

Hypothesis 1 suggests that auditors are inclined to include goodwill in the KAMs for clients with excess goodwill to alleviate their perceived liability. It can be expected that when a firm is in high information transparency, investors and other external stakeholders have a better understanding of the firm's risks and can more easily identify its misreporting. This results in a stronger litigation risk faced by its auditors. Therefore, during the audit process, auditors would be more cautious and more inclined to make KAM disclosures when identifying the client's excess goodwill. We use the information disclosure ratings of listed firms issued by the Shenzhen Stock Exchange to measure information transparency. The information disclosures of listed firms are graded from high to low using the letters A, B, C, and D. We code them as 4, 3, 2, and 1, respectively. We then divide the sample into two groups based on the industry-year median of firms' information disclosure rating. Firms with an information disclosure rating above the median are classified as high information transparency group and others are classified as low

information transparency group.

The subsample regression results for models (1) and (3) are reported in Table 8. Columns (1) and (2) show that the positive effect of excess goodwill on the probability of including goodwill in KAMs in the high information transparency group is significantly larger than that in the low information transparency group, consistent with our expectation. Columns (3) and (4) show no significant difference in the relationship between goodwill KAM disclosures and the probability of goodwill impairment among the two groups.

5.1.4. Audit firm size

We then test whether Hypothesis 1 and Hypothesis 2 exhibit differences across audit firms of varying sizes. It is generally believed that large audit firms, with their “Deep Pockets”, attach more importance to reputational risk (DeAngelo, 1981; Lennox, 1999), and therefore are more likely to remain cautious in goodwill auditing. Additionally, large audit firms typically have more resources and professional expertise to identify and evaluate complex matters related to goodwill, thus they are more likely to recognize excessive goodwill in clients and report goodwill more frequently in KAMs. However, on the other hand, due to the limited volume of business in smaller audit firms, the potential litigation

risks, and reputational losses they face may have a greater impact on their operations. Therefore, they are also very likely to report goodwill in KAMs to mitigate these risks. Additionally, clients of large audit firms may differ from those of smaller firms, with the former having higher pre-audit earnings quality (Chen et al., 2022). As a result, the likelihood of auditors reporting goodwill in KAMs may not be higher for large audit firms compared to smaller ones. For Hypothesis 2, regardless of whether the audit is conducted by a large or small audit firm, the inclusion of goodwill in KAMs indicates that auditors have performed sufficient audit procedures on matters related to goodwill, and they will attract the attention of investors and other stakeholders, thereby influencing the management’s provision for goodwill impairment.

We test Hypothesis 1 and Hypothesis 2 using subsamples audited by Big 4 and non-Big 4 audit firms.⁶ The results are shown in Table 9. Columns (1) and (2) indicate that the coefficient on *GWexcess* is significantly positive in both groups. The magnitude of the coefficient in the Big 4 group is larger than that in the non-Big 4 group, while the difference is not statistically significant. Columns (3) and (4) show that the coefficient on *KAM* is significantly positive in both groups. These results support the notion that both large and small audit firms are capable of

Table 8

Cross sectional analyses: Firms’ information transparency.

	<i>KAM_{it}</i>		<i>IMP_{it}</i>	
	(1)	(2)	(3)	(4)
	Low information transparency	High information transparency	Low information transparency	High information transparency
<i>GWexcess_{it-1}</i>	17.835*** (13.18)	26.016*** (7.01)		
<i>KAM_{it}</i>			0.999*** (9.96)	1.023*** (4.89)
<i>Size_{it}</i>	−0.048 (−0.53)	−0.388** (−2.07)	0.159** (2.10)	0.279* (1.79)
<i>Lev_{it}</i>	0.126 (0.32)	−0.234 (−0.22)	−1.001*** (−3.03)	−1.041 (−1.27)
<i>ROA_{it}</i>	−2.336*** (−2.78)	−0.386 (−0.10)	−9.821*** (−11.37)	−6.935** (−2.51)
<i>Growth_{it}</i>	0.482*** (2.98)	0.174 (0.40)	−0.519*** (−3.71)	−0.083 (−0.23)
<i>Age_{it}</i>	0.073 (0.62)	−0.112 (−0.54)	−0.070 (−0.69)	0.130 (0.67)
<i>First_{it}</i>	−0.016*** (−3.15)	−0.004 (−0.46)	0.005 (1.26)	0.013* (1.76)
<i>Dual_{it}</i>	0.330*** (2.74)	0.771*** (3.04)	0.208** (2.11)	−0.037 (−0.18)
<i>Sep_{it}</i>	0.909 (1.04)	1.686 (1.08)	−0.818 (−1.14)	−0.744 (−0.55)
<i>NTC_{it}</i>	−0.580*** (−2.79)	−0.650 (−1.11)	0.016 (0.10)	0.368 (0.89)
<i>PPE_{it}</i>	−0.574 (−1.13)	−0.870 (−0.83)	−0.289 (−0.66)	−0.663 (−0.79)
<i>SOE_{it}</i>	−0.180 (−1.14)	−0.623** (−2.02)	−0.097 (−0.69)	−0.217 (−0.89)
<i>Big4_{it}</i>	−0.311 (−0.85)	0.385 (0.73)	−0.813** (−2.30)	−1.022** (−2.41)
<i>Cost_{it}</i>	0.373*** (2.68)	0.666** (2.41)	0.171 (1.41)	0.099 (0.46)
<i>Atype_{it}</i>	−0.374 (−1.60)		0.036 (0.16)	
<i>Year</i>	Yes	Yes	Yes	Yes
<i>Ind</i>	Yes	Yes	Yes	Yes
<i>N</i>	3445	884	3453	879
<i>Pseudo R²</i>	0.293	0.366	0.164	0.084
<i>Suest test (Chi²)</i>	4.66**		0.01	

Table 8 presents the cross sectional analysis results for H1 and H2 distinguishing firm’s information transparency. The figures in bold are the variables of our interest. This table presents the coefficients and z-values. ***, **, and * indicate significance at 1 %, 5 %, and 10 %, respectively.

Table 9

Cross sectional analyses: Audit firm size.

	<i>KAM_{it}</i>		<i>IMP_{it}</i>	
	(1)	(2)	(3)	(4)
	Non-Big4	Big4	Non-Big4	Big4
<i>GWexcess_{it-1}</i>	21.302*** (16.24)	26.990*** (5.18)		
<i>KAM_{it}</i>			1.017*** (13.10)	0.937*** (2.82)
<i>Size_{it}</i>	−0.178** (−2.57)	−0.685** (−2.38)	0.117** (2.07)	0.040 (0.18)
<i>Lev_{it}</i>	0.164 (0.52)	0.400 (0.27)	−0.652** (−2.54)	−0.208 (−0.15)
<i>ROA_{it}</i>	−2.445*** (−3.23)	−5.289 (−1.21)	−9.427*** (−13.10)	−6.882** (−1.96)
<i>Growth_{it}</i>	0.471*** (3.74)	2.542*** (4.34)	−0.429*** (−3.93)	−0.878** (−2.02)
<i>Age_{it}</i>	0.015 (0.18)	0.399 (1.48)	0.038 (0.55)	−0.000 (−0.00)
<i>First_{it}</i>	−0.014*** (−3.62)	−0.005 (−0.47)	0.006* (1.87)	0.002 (0.20)
<i>Dual_{it}</i>	0.344*** (3.48)	0.384 (0.80)	0.145* (1.83)	−0.088 (−0.22)
<i>Sep_{it}</i>	0.805 (1.22)	0.255 (0.11)	−0.888* (−1.65)	0.871 (0.45)
<i>NTC_{it}</i>	−0.463*** (−2.77)	0.682 (0.83)	−0.018 (−0.14)	0.585 (0.64)
<i>PPE_{it}</i>	−0.543 (−1.39)	−1.417 (−0.93)	−0.506 (−1.57)	−2.773** (−2.10)
<i>SOE_{it}</i>	−0.223* (−1.84)	−0.305 (−0.67)	−0.152 (−1.51)	0.070 (0.18)
<i>Cost_{it}</i>	0.391*** (3.56)	0.980** (2.08)	0.216** (2.43)	0.187 (0.52)
<i>Atype_{it}</i>	−0.177 (−0.77)	−0.517 (−0.67)	0.084 (0.39)	−0.679 (−0.73)
<i>Year</i>	Yes	Yes	Yes	Yes
<i>Ind</i>	Yes	Yes	Yes	Yes
<i>N</i>	6119	444	6119	455
<i>Pseudo R²</i>	0.295	0.308	0.140	0.121
<i>Suest test (Chi²)</i>	1.13		0.06	

Table 9 presents the cross sectional analysis results for H1 and H2 distinguishing audit firm size. The figures in bold are the variables of our interest. This table presents the coefficients and z-values. ***, **, and * indicate significance at 1 %, 5 %, and 10 %, respectively.

⁶ We also test these hypotheses within subsamples audited by Big 10 and non-Big 10 audit firms, the results are similar (untabulated).

identifying clients' excessive goodwill and signaling goodwill risk in the KAMs, thereby prompting clients to recognize goodwill impairment.

5.2. Further analyses

5.2.1. Auditors' attention to firms' "big bath" through goodwill impairment

Our evidence shows that auditors identify clients' excessive goodwill, make appropriate disclosures in KAMs, and promote goodwill impairment by clients. The basic idea is that managers tend to avoid or delay recognizing goodwill impairment losses. However, there may exist another scenario where managers may engage in opportunistic "big bath" behavior by providing for significant goodwill impairment. In this section, we examine whether auditors pay attention to clients' "big bath" through large goodwill impairment.

We first conduct a univariate test focusing on firms with goodwill impairment. If auditors pay special attention to clients' "big bath" incentives through large goodwill impairment, we expect firms with goodwill as a KAM ($KAM = 1$) have smaller excess goodwill (GW_{excess}).

Table 10
Auditors' attention to firms' "big bath" through goodwill impairment.

	$KAM_{it} = 0$ ($N = 857$)	$KAM_{it} = 1$ ($N = 1556$)	Difference (t-test)
Panel A: t-test on the means of $GW_{excess_{it-1}}$			
$GW_{excess_{it-1}}$	Mean −0.0338	Mean 0.0544	−0.0882***
Panel B: Auditors' attention to firms' "big bath"			
	KAM_{it} (1)	(2)	(3)
$Bath1_{it}$	−0.805*** (−4.46)		
$Bath2_{it}$		−0.388*** (−2.87)	
$Bath3_{it}$			0.152 (0.85)
$Size_{it}$	−0.162** (−2.50)	−0.167** (−2.57)	−0.160** (−2.46)
Lev_{it}	−1.205*** (−4.15)	−1.129*** (−3.88)	−1.115*** (−3.81)
ROA_{it}	−6.871*** (−9.03)	−6.288*** (−8.69)	−4.948*** (−5.89)
$Growth_{it}$	0.611*** (5.96)	0.612*** (5.97)	0.634*** (6.16)
Age_{it}	0.121 (1.53)	0.129 (1.63)	0.125 (1.59)
$First_{it}$	−0.017*** (−4.67)	−0.017*** (−4.70)	−0.017*** (−4.76)
$Dual_{it}$	0.154* (1.68)	0.155* (1.69)	0.152* (1.66)
Sep_{it}	0.800 (1.25)	0.762 (1.19)	0.786 (1.23)
NTC_{it}	−0.672*** (−4.10)	−0.667*** (−4.05)	−0.674*** (−4.10)
PPE_{it}	−2.336*** (−6.06)	−2.304*** (−5.97)	−2.298*** (−5.97)
SOE_{it}	−0.483*** (−4.06)	−0.476*** (−4.00)	−0.470*** (−3.96)
$Big4_{it}$	0.266 (1.33)	0.269 (1.35)	0.257 (1.29)
$Cost_{it}$	0.593*** (5.88)	0.590*** (5.85)	0.590*** (5.84)
$Atype_{it}$	−0.336* (−1.68)	−0.337* (−1.67)	−0.373* (−1.84)
$Year$	Yes	Yes	Yes
Ind	Yes	Yes	Yes
N	6581	6581	6581
$Pseudo R^2$	0.114	0.113	0.112

Table 10 presents the tests on auditors' attention to firms' "big bath" through goodwill impairment. Panel A shows the result of univariate test. Panel B shows the result of regression analysis. The figures in bold are the variables of our interest. This table presents the coefficients and z-values or t-values. ***, **, and * indicate significance at 1 %, 5 %, and 10 %, respectively.

The univariate test results are shown in Panel A of Table 10. Among firms with goodwill impairment, those with goodwill listed as a KAM significantly have larger excess goodwill (GW_{excess}). The result indicates that auditors pay attention to firms' excess goodwill, rather than "big bath" through excessive impairment provision.

Additionally, we conduct a regression analysis. Referring to AbuGhazaleh et al. (2011) and Riedl (2004), this paper sets three variables, $Bath1$, $Bath2$, and $Bath3$, to represent the extent to which a firm uses goodwill impairment for "big bath" accounting. Specifically, if the earnings before goodwill impairment divided by the year-end total assets is negative and the change in firm's pre-impairment earnings to asset ratio from period t-1 to t is less than the median of non-zero negative values, the firm is considered to have a "big bath" motive, and $Bath1$ equals 1; otherwise, it equals 0. If the operating profit before goodwill impairment is negative and the firm also recognizes goodwill impairment, it is considered to have a "big bath" motive, and $Bath2$ equals 1; otherwise, it equals 0. Additionally, if the return on assets (ROA) is negative and the change in ROA from the previous year is less than the median of all non-zero negative values, the firm is thought to have a "big bath" incentive, and $Bath3$ equals 1; otherwise, it equals 0. If auditors pay attention to clients' "big bath" incentives through goodwill impairment, $Bath1$, $Bath2$, and $Bath3$ should be significantly positively correlated with the probability of including goodwill in KAMs. The regression results with $Bath1$, $Bath2$, and $Bath3$ as the independent variables and with KAM as the dependent variable are shown in Panel B of Table 10. The coefficients on $Bath1$ and $Bath2$ are all significantly negative, and the coefficient on $Bath3$ is not significant. These results are inconsistent with the expectation that auditors focus on clients' "big bath" behavior through goodwill impairment. Although "big bath" is also a form of financial statement manipulation, it reduces the firm's profits, which may present lower audit risk for auditors. Thus, auditors focus more on the greater risk of untimely goodwill impairment, rather than excessive impairment.

5.2.2. The effects of a goodwill KAM on clients' future goodwill impairment

Our Hypothesis 2 focuses on the effect of including goodwill in KAMs on clients' current goodwill impairment behavior. We further consider its effect on clients' future goodwill impairment. The auditors' consideration of goodwill as a KAM may imply that there is a certain risk of goodwill, which can be an "early warning" of clients' future goodwill impairment. Additionally, the disclosure of a goodwill KAM can raise stakeholder concerns, which in turn may prompt the firm to recognize goodwill impairment in subsequent periods. We conduct regressions with whether firms recognize goodwill impairment in the following one to three years (IMP_1 , IMP_2 , IMP_3) as the dependent variables and whether there exists a goodwill KAM in the current period (KAM) as the independent variable. The results are shown in columns (1) to (3) of Table 11. We also construct the differences between the excess goodwill in each of the subsequent three periods and the excess goodwill in the previous period ($D_GW_{excess1}$, $D_GW_{excess2}$, $D_GW_{excess3}$) to test whether the inclusion of goodwill in KAMs can contribute to the reduction of excess goodwill in the future periods, and the results are shown in columns (4) to (6) of Table 11.

In columns (1) to (3), the coefficients on IMP_1 , IMP_2 , and IMP_3 are all significantly positive, indicating that the inclusion of goodwill as a KAM significantly contributes to the impairment of goodwill in the next three periods. In columns (4) to (6), the coefficients on $D_GW_{excess1}$, $D_GW_{excess2}$, and $D_GW_{excess3}$ are significantly negative, which indicates that the KAMs related to goodwill promote the firms to reduce their excess goodwill in the following three years. These results underscore the auditors' role in facilitating the timely and adequate recognition of goodwill impairment through KAM disclosures.

6. Conclusion

Stakeholders are concerned that the current goodwill accounting

Table 11

The effects of a goodwill KAM on clients' future goodwill impairment.

	<i>IMP_1</i>	<i>IMP_2</i>	<i>IMP_3</i>	<i>D_GWexcess1</i>	<i>D_GWexcess2</i>	<i>D_GWexcess3</i>
	(1)	(2)	(3)	(4)	(5)	(6)
<i>KAM_{it}</i>	0.980*** (11.99)	0.868*** (8.72)	0.925*** (6.79)	−0.019*** (−12.96)	−0.014*** (−7.66)	−0.012*** (−3.81)
<i>Size_{it}</i>	0.035 (0.60)	0.038 (0.52)	0.013 (0.12)	−0.002 (−1.50)	0.001 (0.58)	−0.003 (−1.17)
<i>Lev_{it}</i>	0.251 (0.93)	0.152 (0.45)	0.068 (0.14)	0.028*** (4.84)	0.037*** (4.93)	0.044*** (3.45)
<i>ROA_{it}</i>	−0.406 (−0.80)	−1.058 (−1.41)	−0.637 (−0.40)	−0.048*** (−3.04)	0.023 (0.99)	0.093* (1.95)
<i>Growth_{it}</i>	−0.353*** (−3.18)	0.008 (0.06)	0.321* (1.69)	0.035*** (9.53)	−0.011** (−2.54)	−0.017*** (−2.93)
<i>Age_{it}</i>	0.090 (1.24)	0.085 (0.96)	−0.067 (−0.57)	0.003** (2.35)	0.002 (1.56)	0.006** (2.55)
<i>First_{it}</i>	0.001 (0.26)	0.001 (0.28)	0.001 (0.17)	0.000*** (4.02)	0.000*** (2.86)	0.000** (2.36)
<i>Dual_{it}</i>	0.166* (1.96)	0.096 (0.93)	0.034 (0.23)	0.003 (1.32)	0.001 (0.39)	0.001 (0.15)
<i>Sep_{it}</i>	−0.250 (−0.45)	−0.020 (−0.03)	−1.431 (−1.49)	−0.001 (−0.13)	−0.019 (−1.37)	−0.000 (−0.01)
<i>NTC_{it}</i>	−0.323** (−2.18)	−0.407** (−2.26)	−0.228 (−0.79)	0.010*** (3.19)	0.006 (1.46)	0.002 (0.31)
<i>PPE_{it}</i>	−1.084*** (−3.21)	−1.309*** (−3.10)	−1.241** (−2.20)	0.019*** (3.54)	0.005 (0.82)	0.032*** (2.87)
<i>SOE_{it}</i>	−0.203* (−1.88)	−0.201 (−1.53)	−0.293 (−1.62)	0.004** (2.41)	0.001 (0.56)	−0.002 (−0.55)
<i>Big4_{it}</i>	−0.796*** (−4.01)	−0.577** (−2.40)	−0.634** (−2.00)	0.010*** (4.27)	0.007** (2.54)	0.006 (1.45)
<i>Cost_{it}</i>	0.260*** (2.82)	0.204* (1.84)	0.248* (1.65)	−0.003** (−1.97)	−0.005** (−2.52)	−0.000 (−0.11)
<i>Atype_{it}</i>	0.271 (1.21)	−0.335 (−1.10)	0.102 (0.20)	0.002 (0.42)	−0.002 (−0.42)	0.005 (0.32)
<i>Constant</i>	−5.048*** (−4.37)	−4.048*** (−2.89)	−4.422** (−2.39)	0.055*** (3.15)	0.047** (2.21)	0.049 (1.57)
<i>Year</i>	Yes	Yes	Yes	Yes	Yes	Yes
<i>Ind</i>	Yes	Yes	Yes	Yes	Yes	Yes
<i>N</i>	4506	2701	1207	4506	2701	1210
<i>Adjusted R²</i>				0.103	0.068	0.080
<i>Pseudo R²</i>	0.076	0.066	0.082			

Table 11 presents the effects of a goodwill KAM on clients' future goodwill impairment. The figures in bold are the variables of our interest. This table presents the coefficients and z-values or t-values. ***, **, and * indicate significance at 1 %, 5 %, and 10 %, respectively.

with the “impairment-only approach” may result in untimely impairments, which the IASB believes is best addressed by auditors and regulators instead of by amending the standard itself. In light of this, our study aims to explore the effectiveness of auditing in identifying inflated goodwill among clients and encouraging the prompt recognition of goodwill impairment. We delve into the content of KAMs and use a sample of Chinese listed firms to provide empirical evidence. We find that firms' excess goodwill is positively associated with the probability that auditors identify goodwill as a KAM; and the inclusion of goodwill in KAMs prompts clients to recognize goodwill impairment. The former association is notably stronger for firms receiving clean audit opinions, having small profits before goodwill impairment recognition, or having high information transparency, and the latter effect is mitigated for firms with small profits. Both effects hold in large and small audit firms with no significant difference. Further analyses indicate that auditors exhibit little concern regarding clients' “big bath” incentives via goodwill impairments. Additionally, firms with goodwill communicated as a KAM are more likely to recognize goodwill impairments in future periods and continuously reduce their excess goodwill. After considering endogeneity issues and using alternative measures for key variables, the main findings of this study still hold.

Drawing from the analyses of KAMs, this study demonstrates that auditors can identify and disclose clients' excess goodwill based on a cost-benefit analysis, which in turn encourages management to

recognize goodwill impairment. It indicates that auditors can help correct management' opportunistic behavior and irrational biases in goodwill impairment recognition. This empirically supports IASB's assertion that effective implementation of goodwill impairment requires audit and regulatory safeguards. Further, the results of this study shed light on the practical significance relating to the disclosure of KAMs under the new auditing standards. When an account is identified as a KAM, it not only alerts the external stakeholders of the possible risks of the item, but also exerts pressure on management to raise their behavioral standards, thereby enhancing the quality of accounting information. This study enriches the existing literature on KAMs and goodwill impairment, offering significant practical implications.

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Data availability

The data that support the findings of this study are available from the authors upon reasonable request.

Appendix 1

Description of variables.

Variables	Description
KAM_{it}	An indicator variable that equals 1 if “goodwill” is included in KAMs of firm i in year t , and 0 otherwise.
IMP_{it}	An indicator variable that equals 1 if goodwill is impaired for firm i in year t , and 0 otherwise.
$GW_{excess_{it-1}}$	The excess goodwill of firm i at the end of year $t-1$. The excess goodwill of a firm is measured as the regression residual of model (2) (a goodwill expectation model based on Luo and Guo (2023) and Wei and Zhu (2019)).
$Size_{it}$	The natural log of total assets.
Lev_{it}	The closing balance of the total liabilities divided by total assets.
ROA_{it}	Total net income divided by total assets.
$Growth_{it}$	The growth rate of operating income.
Age_{it}	The natural log of the number of years the company has been listed.
$First_{it}$	The percentage of shares of the largest shareholder.
$Dual_{it}$	An indicator variable that equals 1 if there is a duality of CEO and chairman, and 0 otherwise.
Sep_{it}	The difference between the ownership proportion (cash flow rights) and the control proportion (voting rights) of the actual controlling shareholder.
NTC_{it}	The difference between the sum of accounts payable and accounts collected in advance and the sum of accounts receivable and prepaid accounts, divided by operating income.
PPE_{it}	The ratio of net fixed assets to total assets.
SOE_{it}	An indicator variable that equals 1 if the firm is state-owned, and 0 otherwise.
$Big4_{it}$	An indicator variable that equals 1 if the audit firm is an international Big 4, and 0 otherwise.
$Cost_{it}$	The natural log of the audit fees.
$Atype_{it}$	An indicator variable that equals 1 if the audit opinion is a modified audit opinion, and 0 otherwise.

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